

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Computer Science and Design) SEMESTER - 3 Winter 2025 (Regular)

Course :Bachelor of Technology (Computer Science and Design) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: 25AF1257PC302 - FUNDAMENTALS OF DATA STRUCTURES

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Which of the following is a linear data structure?
 1. Tree
 2. Graph
 3. Stack
 4. Binary Tree
 - 2 The time complexity of binary search algorithm is:
 1. $O(n)$
 2. $O(\log n)$
 3. $O(n^2)$
 4. $O(n \log n)$
 - 3 In a circular queue, what happens when rear reaches the end of the array?
 1. Queue becomes full
 2. Rear wraps around to the beginning
 3. Queue overflow occurs
 4. Elements are deleted
 - 4 Which notation does not require parentheses for expression evaluation?
 1. Infix
 2. Prefix
 3. Postfix
 4. Both b and c
 - 5 The best case time complexity of bubble sort is:
 1. $O(n)$
 2. $O(n^2)$
 3. $O(\log n)$
 4. $O(n \log n)$

- 6 What is the difference between static and dynamic memory allocation?
1. Static is at compile time, dynamic is at runtime
 2. Static uses heap, dynamic uses stack
 3. No difference
 4. Static is slower than dynamic
- 7 In a doubly linked list, each node contains:
1. Only data
 2. Data and one pointer
 3. Data and two pointers
 4. Only pointers
- 8 Which data structure is used for recursion?
1. Queue
 2. Stack
 3. Array
 4. Linked List
- 9 What is an Abstract Data Type (ADT)?
1. A physical implementation of data
 2. A logical description of data and operations
 3. A programming language
 4. A sorting algorithm
- 10 In row major order, elements of a 2D array are stored:
1. Column by column
 2. Row by row
 3. Diagonally
 4. Randomly
- 11 The operation of removing an element from the stack is called:
1. Push
 2. Pop
 3. Peek
 4. Delete
- 12 Which of the following sorting algorithms is most efficient for small datasets?
1. Quick Sort
 2. Merge Sort
 3. Insertion Sort
 4. Heap Sort

Q2. Solve the following.

- A) Explain the characteristics of an algorithm. Differentiate between algorithm and program with suitable examples. 6
- B) What is an Abstract Data Type (ADT)? Explain Array as an ADT with its operations. 6

Q3. Solve the following.

- A) Explain linear search and binary search algorithms. Compare their time complexities. Write an algorithm for binary search. 6

- B) Given an array: [64, 34, 25, 12, 22, 11, 90]. Perform bubble sort and show the array after each pass. Also calculate the number of comparisons made. 6
- Q4. Solve Any Two of the following.
- A) What is a sparse matrix? Explain how sparse matrices are represented using arrays. 6
- B) Explain the concept of polynomial representation using arrays. Add the following polynomials: 6
- $$P(x) = 5x^4 + 3x^2 + 2$$
- $$Q(x) = 4x^3 + 3x^2 + 5x + 1$$
- C) Explain the address calculation for 2D arrays in both row major and column major order. For a 2D array A[10][20] with base address 1000 and each element occupying 4 bytes, calculate the address of element A[5][10] using row major order. 6
- Q5. Solve Any Two of the following.
- A) What is a linked list? Explain the advantages of linked lists over arrays. Describe the operations performed on a singly linked list with suitable diagrams. 6
- B) Differentiate between: Singly linked list and Doubly linked list. Write an algorithm to insert a node at the beginning of a singly linked list. 6
- C) Explain dynamic memory allocation. Write algorithms to create and traverse a singly linked list. Show with an example how nodes are linked using pointers. 6
- Q6. Solve Any Two of the following.
- A) What is a stack? Explain stack operations (Push and Pop) with diagrams. Write algorithms for Push and Pop operations on a stack implemented using arrays. 6
- B) Convert the following infix expression to postfix notation and evaluate the postfix expression: 6
- $$\text{Infix: } (A + B) * C - (D - E) / F$$
- where A=5, B=3, C=4, D=10, E=2, F=2
- Explain the steps clearly.
- C) What is a queue? Explain the concept of circular queue and its advantages over linear queue. Write an algorithm to insert an element in a circular queue. Show with an example how circular queue overcomes the limitations of linear queue. 6

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